Question-1:

Convert 123.45689_{10} into 18 bit IEEE754 floating point format where the size of the fraction field is 9 bits.
Note: consider 6 floating points while converting from decimal to binary.

Question-2:

Find the corresponding decimal number of this 0xAB25 16 bit IEEE754 floating point format where the size of the fraction field is 9 bits.
Note: consider 6 floating points during conversion.

Question-3:

$1101010.11010 \times 2^{245}$; Can we format/convert this number into IEEE754 21 bit representation where size of the fraction field is 15 bits?

Question-4:

What is the range of the exponent field for a 20-bit IEEE754 system where the size of the fraction field is 12 bits.

Question-5:

For a 20-bit register with 1 bit for sign, 6 bits for exponents, and the rest for fractions, what will be the actual exponent range of the numbers that can be represented using IEEE 754 floating point representations?

Question-6:

Convert 123.45689_{10} into IEEE754 double precision floating point format.

Note: consider 6 floating points while converting from decimal to binary.